An extremal even unimodular lattice in dimension 112

Gluing with $\mathrm{SL}_2(29)$ symmetry and a computer-assisted proof

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Abstract

We construct an explicit positive-definite even unimodular lattice L_{112} of dimension 112 and minimum norm 10, attaining the extremal bound. The construction glues two copies of a 56-dimensional extremal lattice of Nebe using an endomorphism whose square is -5 and which commutes with $\mathrm{SL}_2(29)$. Extremality is reduced to finitely many coset bounds and proved by checking orbit completeness and either canonical forms or integer conditions derived from the midpoint lemma. We also determine the full automorphism group, of order $1\,169\,280$. Defining matrices, verification data and code are supplied, and the deduction from explicit computational hypotheses is formalized in Lean.

1. Introduction

1.1 Extremal even unimodular lattices

A positive-definite lattice of rank n is a discrete subgroup $\Lambda = \mathbb{Z}b_1 \oplus \ldots \oplus \mathbb{Z}b_n$ spanning Euclidean space $\mathbb{R}^n$. We write $\langle x, y\rangle$ for the inner product and call $\langle x, x\rangle$ the norm of x. The lattice is even if every norm is an even integer. An integral lattice is unimodular if its Gram matrix has determinant 1, or equivalently if it equals its dual. Even unimodular lattices exist only in dimensions divisible by 8, and an example exists in every such positive dimension [SPLAG, Ch. 7].

The theta series $\theta_{\Lambda(q)} = \sum_{x\in\Lambda} q^{\langle x,x\rangle/2}$ of an even unimodular lattice is a modular form of weight $\frac{n}{2}$. This gives the bound of Mallows–Odlyzko–Sloane [MOS75]:

$$\min \Lambda \le \mu(n) := 2\left\lfloor \frac{n}{24} \right\rfloor + 2. \tag{1.1}$$

Here $\min\Lambda$ is the smallest norm of a nonzero vector. An even unimodular lattice is extremal when equality holds in (1.1). Unless another class of lattices is explicitly specified, "extremal lattice" in this paper means extremal even unimodular lattice. We write $\mu(n)$ for this bound.

The bound is 6 in dimension 56 and 10 in dimension 112. Thus the construction below must exclude nonzero vectors of norms 2, 4, 6 and 8 in dimension 112. Examples in lower dimensions include E_8 and the Leech lattice. More recently, Sakamoto's preprint [Sak26] constructs two examples in dimension 88. Further background and comparison tables are collected in Appendix G; the proof of the result below does not depend on those comparisons.

1.2 Construction and main result

Let L_{56} denote the 56-dimensional lattice whose Gram matrix is the file `L56_gram.txt` listed in Appendix D. (The defining matrices are not printed in this paper; they are distributed as data files identified

by their SHA-256 hashes. Appendix D also records a specification of L_{56} and τ in terms of Nebe's construction that does not depend on the files.) The lattice L_{56} is $\mathrm{SL}_2(29)$-equivariantly isometric to Nebe's lattice $L_{56,2}\big(\widetilde{M}\big)$ [Neb98, Table II]; the identification is established in §7.2. The two matrices in the file `sl2_29_generators.txt` are isometries of L_{56} and generate a group S isomorphic to $\mathrm{SL}_2(29)$ (F4). Let τ be the endomorphism given by the file `tau.txt`. It commutes with S and satisfies

$$\tau^2 = -5\,\mathrm{id}, \quad \langle x\tau, y\tau\rangle = 5\langle x, y\rangle \quad (x, y \in L_{56}).$$

Matrices act on row vectors on the right. These identities and the defining matrices are checked by [C2]. For this choice of τ, set

$$L_{112} = L_\tau := \left\{ \frac{(x, x\tau + 2y)}{\sqrt{2}} : x, y \in L_{56} \right\} \subset \frac{1}{\sqrt{2}}(L_{56} \perp L_{56}). \tag{1.2}$$

Theorem A. The lattice L_{112} defined by the data files of Appendix D and (1.2) is an even unimodular lattice of dimension 112 with minimum norm 10. In particular, it is extremal.

The diagonal action of S preserves L_{112}. The proof of Theorem A is outlined in §1.3 and completed in §4.3 using the finite computations in §6. The construction and the verification are given separately, so the verification does not require repeating the search that led to τ.

The choice of τ is not unique. The centralizer C of S in $\mathrm{Aut}(L_{56})$ has order 12, and exactly 24 endomorphisms of L_{56} commute with S and satisfy $\tau^2 = -5$ [C2]. Only the reduction of τ modulo $2L_{56}$ enters the glued lattice. Proposition 3.3 classifies these reductions into three classes of gluing maps; the 24 endomorphisms reduce into two of them, called B and C below, 12 into each [C2], and endomorphisms in the same class give isometric glued lattices. The supplied τ lies in class C, which gives L_{112}. Class B gives an even unimodular lattice with minimum norm 8 and exactly 292 320 vectors of norm 8 [C4]. This second lattice serves as a positive control: the verification programs must detect its short vectors. The classification of the gluings is completed in §3.3.

Two further conclusions are treated separately from the extremality proof. The extremal theta series determines that L_{112} has 10 888 335 360 vectors of norm 10 and 247 006 775 232 000 vectors of norm 12 (Appendix B). Its full automorphism group has order 1 169 280 (Theorem B, §8 and Appendix F). Theorem A does not use the determination of this full group.

1.3 Outline of the proof and the verification methods

We first verify that the lattice in (1.2) is even unimodular (Lemma 3.2). The main remaining task is to prove that it has no nonzero vector of norm below 10. The minimum norm of L_{56} is 6; this is checked by two rigorous enumerations, one using interval arithmetic and one using an integer quadratic form (§6.1). Consequently, a nonzero vector $\frac{(x,v)}{\sqrt{2}}$ of norm below 10 in L_{112} would have both components in L_{56} and would satisfy

$$v \equiv x\tau \pmod{2L_{56}}, \quad (\langle x, x\rangle, \langle v, v\rangle) \in \{(6,6), (6,10), (10,6), (8,8)\}.$$

Lemma 4.2 shows that this list is exhaustive and that neither component is zero. Since $\tau^2 \equiv 1 \pmod 2$, interchanging the two components reduces the type $(10, 6)$ to $(6, 10)$ (Appendix C.4). It therefore suffices to establish the following two bounds:

$$\min(r\tau + 2L_{56}) \geq 14 \quad \text{for all } r \in L_{56}(6),$$

$$\min(x\tau + 2L_{56}) \geq 12 \quad \text{for all } x \in L_{56}(8),$$

where $L_{56}(m)$ denotes the vectors of norm m in L_{56} and the minimum of a coset is taken over all its vectors. This is Corollary 4.3.

Because the group S preserves norms and commutes with τ, each bound need only be checked on a collection of orbit representatives. For norm 6 there are 656 orbits. For norm 8 the supplied collections give 1 527 984 test vectors of L_{56}, with repetitions, covering 1 517 130 distinct orbits. These are vectors of the 56-dimensional lattice, not norm-8 vectors of L_{112}. The completeness check verifies the norms, disjointness and orbit sizes, and compares the total sizes with the shell sizes determined by the theta series (§§6.2–6.4).

The lower bounds are then verified as follows. For each norm-6 representative, we supply a vector in its prescribed coset whose norm and inner products with all minimal vectors of L_{56} imply the bound 14. The implication follows from the midpoint lemma and Corollary 5.2; the supplied vector and its conditions are checked in §6.6. For the norm-8 test vectors, there are two alternatives. One uses canonical forms to test whether the prescribed coset contains a norm-8 vector (§6.5). The other checks coset representatives whose norm and inner products imply the bound 12 (Corollary 5.3 and §6.7). Either alternative, together with the common norm-6 bounds and completeness checks, proves the required absence of short vectors. The upper bound (1.1) then gives $\min L_{112} = 10$.

Thus the two even-coset tests are alternative ways of completing the same proof, and their shared inputs are part of the verification. Separately written implementations check these inputs and computations. Spherical-design identities, changes of frame and tests on the second gluing provide further consistency checks. Floating-point closest-vector searches and random samples are also reported, but are not substituted for the rigorous checks in the proof. Supplement §S4 describes a further route through support modules. It is not used in the proof; nor is it independent of the proof, since it relies on the same exclusion of type $(8, 8)$ (F5b) as the main argument. The Lean formalization checks the deduction from the stated computational hypotheses (§7.1).

1.4 Relation to previous constructions and the choice of dimension

The construction uses established ingredients: Nebe's 56-dimensional lattices [Neb98], gluing along cosets modulo twice a lattice [Que84], and the Hermitian tensor-product framework used in dimensions 72 and 88 [Neb12, CN13, Sak26]. Proposition 3.4 identifies (1.2) with a Hermitian tensor product over $\mathbb{Z}\left[\sqrt{-5}\right]$. The result of this paper is the extremality of the lattice obtained with the supplied endomorphism, together with the explicit data and verification needed to establish it.

The choice of dimension 112 was motivated by comparison with dimensions 96 and 104, where the extremal bound is also 10. The Siegel weighted mean of the number of nonzero vectors of norm below 10 is approximately 15 732 086, 1 046 569 and 51 014 in these three dimensions, respectively [C1]. The smaller mean in dimension 112 suggested a useful direction for the search. These are averages over a genus, not estimates of the time needed to prove extremality. The available $\mathrm{SL}_2(29)$ action on L_{56} also supplies the symmetry used to reduce the verification to orbit representatives. Supplement §S1 gives the numerical comparisons and the search motivation.

1.5 Organization of the paper

Sections 2–6 prove Theorem A: notation, construction, reduction to four types, coset bounds and finite computations, respectively. Section 7 describes the scope of the Lean formalization and identifies the base lattice with Nebe's lattice. Section 8 determines the full automorphism group. Appendices A–G

collect terminology, theta series, auxiliary proofs, data and code, pseudocode, the group-theoretic proof and known examples.

A separate supplement contains the Siegel means and search motivation (Supplement §§S1–S2), verification routes, running times and additional tests (Supplement §S3), the support-module argument (Supplement §S4), and a comparison of constructions considered in the search (Supplement §S5). Its sections and tables carry the prefix S.

2. Preliminaries and terminology

We fix conventions for lattices, cosets, group actions and Hermitian forms. Norm always means squared length.

2.1 Conventions and terminology

We use L_{56} for the fixed lattice of §1; the letter L in general definitions and lemmas denotes an arbitrary lattice satisfying the stated assumptions. An isometry between lattices is an additive-group isomorphism preserving the inner product; it need not preserve orientation.

Lattices are given by Gram matrices $G \in \mathbb{Z}^{n\times n}$ with respect to a basis; vectors are integer **row** vectors x, with $\langle x, y\rangle = xGy^T$. A matrix g acts by $x \mapsto xg$; it is an isometry if $gGg^T = G$. "Norm" always means $\langle x, x\rangle$, never its square root. For a lattice L, the shell of norm m is $L(m) = \{x \in L : \langle x, x\rangle = m\}$.

Table 1. Terms distinguished in this paper.

term	symbol	meaning
minimum (of a lattice)	$\min \Lambda$	smallest norm of a nonzero vector of a specific lattice Λ
extremal bound	$\mu(n)$	the upper bound $2\lfloor n/24\rfloor + 2$ for the minimum in dimension n; $\mu(56) = 6$, $\mu(112) = 10$
inner-shell vector	—	nonzero vector of norm $< \mu(n)$; extremal $\iff$ none exists
coset minimum	$\min(c)$	smallest norm of a vector in a coset c modulo $2L$ (§2.2)
centralizer	C	elements of $\mathrm{Aut}(L)$ commuting with every element of $\mathrm{SL}_2(29)$ (§2.4); not the full unit group of an order

2.2 Cosets modulo $2L$

For an even unimodular L, the quotient $L/2L$ is an $\mathbb{F}_2$-vector space of dimension n; we call its elements $c = x + 2L$ **cosets**. Since $\langle x + 2y, x + 2y\rangle = \langle x, x\rangle + 4\langle x, y\rangle + 4\langle y, y\rangle$, the function $q(x + 2L) = \langle x, x\rangle/2 \bmod 2$ is a well-defined quadratic form on $L/2L$. We call a coset *even* or *odd* according to q; all vectors in an even coset have norm $\equiv 0 \pmod 4$, and all vectors in an odd coset have norm $\equiv 2 \pmod 4$. A linear map f on $L/2L$ is **parity-preserving** if $q(cf) = q(c)$ for all cosets. In dimension 56 there are $2^{55} + 2^{27}$ even cosets (including 0) and $2^{55} - 2^{27}$ odd cosets (these are the numbers of zeros and of nonzeros of a nondegenerate quadratic form of plus type).

2.3 Extremal theta series and shells of L

If L is extremal of dimension 56, its theta series is

$$\theta_L = 1 + 15\,590\,400\,q^3 + 36\,957\,286\,800\,q^4 + 15\,284\,192\,071\,680\,q^5 + \cdots$$

[C1] (by [Ser73, Ch. VII], the space of modular forms of weight 28 is 3-dimensional, so "constant term 1, no q, no q^2" determines the form; Appendix B). Thus $|L(6)| = 15\,590\,400$, $|L(8)| = 36\,957\,286\,800$,

$|L(10)| = 15\,284\,192\,071\,680$. Moreover each nonempty shell of an extremal lattice of dimension $24k+8$ is a spherical 7-design (Venkov; see [BV01]); we use this only as a consistency test (§6.3).

2.4 Nebe's lattices with $\mathrm{SL}_2(29)$ and the centralizer

The group $\mathrm{SL}_2(29)$ has a faithful irreducible representation of degree 14 whose character field is $\mathbb{Q}(\sqrt{29})$ and which has Schur index 2 over $\mathbb{Q}$; as a rational representation it has degree 56, and its endomorphism algebra is the totally definite quaternion algebra $\mathcal{Q}$ over $\mathbb{Q}(\sqrt{29})$ ramified only at the two infinite places [Neb98]. Nebe constructs $\mathrm{SL}_2(29)$-invariant even unimodular lattices $L_{56,2}(M)$ and $L_{56,2}(\widetilde{M})$ attached to the two conjugacy classes of maximal orders $M, \widetilde{M}$ of $\mathcal{Q}$, and reports minimum 6 for both, computed with PARI [Neb98, Table II]. For L_{56} we re-establish the minimum rigorously in §6.1.

For an $\mathrm{SL}_2(29)$-invariant lattice L we write $M' = \mathrm{End}_{\mathrm{SL}_2(29)}(L)$ for the ring of integer matrices commuting with $\mathrm{SL}_2(29)$ and preserving L ($\mathbb{Z}$-rank 8). Its elements that are isometries of L form the **centralizer** $C = C_{\mathrm{Aut}\,L}(\mathrm{SL}_2(29))$. The adjoint defined by the inner product agrees with the standard quaternion involution, so $uu^* = 1$ is equivalent to reduced norm 1. The centralizer C is the finite group of elements of reduced norm 1 of M'; it has order 24 for the catalog lattice $L_{56,2}(M)$ and order 12 (6 up to sign) for the lattice L_{56} used here [C2].

2.5 Hermitian lattices and trace forms

Let $K = \mathbb{Q}(\sqrt{-5})$ with ring of integers $\mathcal{O} = \mathbb{Z}[\sqrt{-5}]$ (class number 2). If τ is an endomorphism of a lattice L with $\tau^2 = -5$ and $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$, then L becomes an $\mathcal{O}$-module P_τ via $\sqrt{-5}\cdot x := x\tau$, and

$$h(x,y) = \langle x, y\rangle + \frac{\langle x\tau, y\rangle}{\sqrt{-5}}$$

is a Hermitian form (linear in the first variable) with $\mathrm{Re}\, h = \langle\cdot,\cdot\rangle$ (Appendix C.1). For Hermitian $\mathcal{O}$-lattices A, B the tensor product $A \otimes_{\mathcal{O}} B$ carries the real form $\mathrm{Re}(h_A \otimes h_B)$; this is the Hermitian tensor product of [CN13].

3. The construction

We glue two copies of a 56-dimensional lattice along a correspondence of cosets modulo $2L$ to obtain a 112-dimensional lattice (§3.1). Listing all gluings that preserve the $\mathrm{SL}_2(29)$-symmetry gives only five cases (§3.3, Table 2): two have inner-shell vectors forced by symmetry, two have minimum 8, and the remaining one is our L_{112}. Finally we rewrite L_{112} as a Hermitian tensor product (§3.4).

3.1 Gluing two copies modulo 2

Definition 3.1 (gluing). Let L be even unimodular of dimension n and f an invertible parity-preserving linear map on $L/2L$. The **gluing** of two copies of L along f is

$$L_f = \left\{ \frac{1}{\sqrt{2}}(x,v) : x, v \in L,\ v + 2L = (x+2L)f \right\}, \qquad \langle \frac{(x,v)}{\sqrt{2}}, \frac{(x,v)}{\sqrt{2}} \rangle = \frac{1}{2}(\langle x, x\rangle + \langle v, v\rangle).$$

We call f the gluing map and L_f the glued lattice. Taking $L = L_{56}$, when f is the reduction of an integer matrix τ with $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$, we get the lattice L_τ of (1.2).

Lemma 3.2. Let L be even unimodular of dimension n and $\tau \in \mathrm{End}(L)$ with $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$. Then L_τ is an even unimodular lattice of dimension $2n$.

Proof. The map $L \oplus L \to L/2L$, $(x, v) \mapsto v - x\tau + 2L$ is a surjective homomorphism (already on $x = 0$). Its kernel L'_τ has index 2^n in $L \oplus L$, so $\det L'_\tau = 2^{2n} \det(L \perp L) = 2^{2n}$, and scaling by $1/\sqrt{2}$ multiplies the $2n$-dimensional Gram determinant by 2^{-2n}; hence $\det L_\tau = 1$. For evenness write $v = x\tau + 2y$:

$$\langle x, x\rangle + \langle v, v\rangle = \langle x, x\rangle + 5\langle x, x\rangle + 4\langle x\tau, y\rangle + 4\langle y, y\rangle = 6\langle x, x\rangle + 4(\langle x\tau, y\rangle + \langle y, y\rangle) \equiv 0 \pmod 4,$$

since $\langle x, x\rangle$ is even. So $\frac{1}{2}(\langle x, x\rangle + \langle v, v\rangle)$ is even. A lattice all of whose norms are even integers is integral by polarization. □

The basis $(e_i, e_i\tau)/\sqrt{2}$, $(0, 2e_i)/\sqrt{2}$ $(1 \le i \le 56)$ of L_τ has Gram matrix

$$\begin{pmatrix} 3G & \tau G \\ (\tau G)^T & 2G \end{pmatrix}, \tag{3.1}$$

which is the form in which L_{112} is distributed (Appendix D).

3.2 Forced collisions from the centralizer

If f agrees modulo 2 with an element $u \in C$, then $(x, xu)/\sqrt{2} \in L_f$ for all x, and a minimal vector x of L_{56} gives an inner-shell vector of norm $\frac{1}{2}(6 + 6) = 6$. We call this a **forced collision**: it is caused by symmetry, not by chance. Examples of such constructions are discussed in Supplement §S5.

3.3 All $\mathrm{SL}_2(29)$-equivariant gluings

To preserve the $\mathrm{SL}_2(29)$-symmetry, we require that f commute with $\mathrm{SL}_2(29)$ acting on $L_{56}/2L_{56}$. The ring of $\mathbb{F}_2$-linear maps on $L_{56}/2L_{56}$ commuting with $\mathrm{SL}_2(29)$ has $\mathbb{F}_2$-dimension 8 and equals the image of $M'/2M'$ [C3], so every commuting map is the reduction of an element of M'. Since 2 is inert in $\mathbb{Q}(\sqrt{29})$ and $\mathcal{Q}$ is unramified at 2, $M'/2M' \cong M_2(\mathbb{F}_4)$.

Proposition 3.3. (a) Exactly 60 elements of $M'/2M'$ are invertible and parity-preserving [C3]. (b) $L_f \cong L_{ufu'}$ for $u, u' \in C$, and $L_f \cong L_{f^{-1}}$. (c) Modulo (b), the 60 maps fall into 2 classes for $L_{56,2}(M)$ and 3 classes for our L_{56} (Table 2) [C3].

For (b): $(x, v) \mapsto (xu, vu')$ maps L_f isometrically onto $L_{u^{-1}fu'}$, and $(x, v) \mapsto (v, x)$ maps L_f onto $L_{f^{-1}}$.

Table 2. The five $\mathrm{SL}_2(29)$-equivariant gluings L_f (f from $M'/2M'$) of two copies of a lattice with $\mathrm{SL}_2(29)$.

name	base	class of f (size)	$\min L_f$	inner-shell vectors
—	$L_{56,2}(M)$	contains an element of C (12)	6	forced collision
A	$L_{56,2}(M)$	does not (48)	8	292 320 of norm 8 [C4]
—	L_{56}	contains an element of C (6)	6	forced collision
B	L_{56}	does not (36)	8	292 320 of norm 8 [C4]
C	L_{56}	does not (18)	**10 (Theorem A)**	**none**

Both surviving classes B and C on L_{56} lift to elements $\tau \in M'$ with $\tau^2 = -5$. Of the 24 such elements, 12 reduce modulo 2 into class B and 12 into class C [C2]. Gluings A and B are isometric: an $\mathrm{SL}_2(29)$-equivariant isometry between them is supplied with the verification data and checked in integer arithmetic [C28]. Thus the two gluings of minimum 8 in Table 2 are the same lattice. More generally, every even unimodular lattice of dimension 112 on which $\mathrm{SL}_2(29)$ acts as on L_{112} is $L_{56,2}(M) \otimes_M \Lambda$ for a unimodular Hermitian lattice Λ of rank 2 over the maximal order M, and there are exactly eight such Λ up to isometry; Table 2 contains four of the resulting lattices. Supplement §S6 states and proves this classification [C29, C30, C31].

3.4 The Hermitian description

Proposition 3.4. Let H be the free Hermitian $\mathcal{O}$-lattice with basis e_1, e_2 and Gram matrix $\begin{pmatrix} 3 & \sqrt{-5} \\ -\sqrt{-5} & 2 \end{pmatrix}$ (so $\det H = 6 - 5 = 1$), and $P = P_\tau$ obtained from L_{56} as in §2.5. Then

$$\varphi : H \otimes_{\mathcal{O}} P \to L_\tau, \qquad e_1 \otimes x + e_2 \otimes y \mapsto \frac{1}{\sqrt{2}}(x,\, x\tau + 2y)$$

is an isometry for the real form $\mathrm{Re}(h_H \otimes h_P)$. (Appendix C.2)

This realizes L_{112} in the Hermitian tensor-product framework of [CN13, Sak26]. Its full automorphism group is determined in §8.

4. Reduction to collisions

We show that an inner-shell vector of L_{112} corresponds to a pair (x, v), with $x \in L_{56}$ and v in the coset $x\tau + 2L_{56}$, whose norms are $(6,6)$, $(6,10)$, $(10,6)$ or $(8,8)$ (§4.1). The absence of such "collisions" can be expressed in terms of coset minima (Corollary 4.3), and the $\mathrm{SL}_2(29)$-symmetry reduces it to finitely many cases (§4.2, Table 3). We then state the computational facts F0–F5 on which the rest of the proof rests and derive Theorem A from them (§4.3).

4.1 Four types

From now on L_{56} is even unimodular of dimension 56 with $\min L_{56} = 6$ and τ satisfies $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$. The adjoint of τ is $\tau^* = 5\tau^{-1}$; since $\tau^2 = -5$, $\tau^* = -\tau$ (Appendix C.1).

Definition 4.1 (collision). For $a, b \in \{6, 8, 10\}$, a **collision of type (a, b)** is a pair (x, v) with $x \in L_{56}(a)$, $v \in L_{56}(b)$ and $v \equiv x\tau \pmod{2L_{56}}$. Equivalently, the coset $x\tau + 2L_{56}$ of some $x \in L_{56}(a)$ contains a vector of norm b.

Lemma 4.2. If $z = \frac{1}{\sqrt{2}}(x, v) \in L_\tau$ is an inner-shell vector ($z \neq 0$, $\langle z, z\rangle \leq 8$), then (x, v) is a collision of type $(6,6)$, $(6,10)$, $(10,6)$ or $(8,8)$.

Proof. If $x = 0$ then $v \in 2L_{56} \setminus \{0\}$, so $\langle v, v\rangle \geq 4 \cdot 6$ and $\langle z, z\rangle \geq 12$. If $v = 0$ then $x\tau \in 2L_{56}$; applying τ^* gives $5x \in 2L_{56}$, hence $x = 5x - 2 \cdot 2x \in 2L_{56}$ and again $\langle z, z\rangle \geq 12$. So $x, v \neq 0$ and $\langle x, x\rangle, \langle v, v\rangle \geq 6$. By the proof of Lemma 3.2, $\langle x, x\rangle + \langle v, v\rangle \equiv 0 \pmod 4$, and $\langle x, x\rangle + \langle v, v\rangle \leq 16$. The only pairs of even integers ≥ 6 with these properties are $(6,6)$ with sum 12 and $(6,10), (8,8), (10,6)$ with sum 16. □

In particular, the type $(6,8)$, which would be the most frequent one by a naive count, is excluded by parity. Since $\tau^2 = -5 \equiv 1 \pmod 2$, the coset map $c \mapsto c\tau$ is an involution on $L_{56}/2L_{56}$, so a collision (x, v) of type $(10,6)$ is the same as a collision (v, x) of type $(6,10)$ (Appendix C.4).

Corollary 4.3. L_τ has no inner-shell vectors if and only if
(i) $\min(r\tau + 2L_{56}) \geq 14$ for every $r \in L_{56}(6)$ (the coset is odd, so its norms are $6, 10, 14, \ldots$; this says 6 and 10 do not occur), and
(ii) $\min(x\tau + 2L_{56}) \geq 12$ for every $x \in L_{56}(8)$ (the coset is even, so its norms are $8, 12, \ldots$; this says 8 does not occur).

(That the cosets in (ii) contain neither 0 nor vectors of norm 4 follows from the case $v = 0$ in the proof of Lemma 4.2 and from $\min L_{56} = 6$.)

4.2 Symmetry

Let $S = \mathrm{SL}_2(29) \subset \mathrm{Aut}(L_{56})$. Every $g \in S$ commutes with τ (exactly, not only modulo 2) [C2]. Hence the coset of $(xg)\tau$ is the image under g of the coset of $x\tau$, and g preserves norms. So condition (i) needs to be tested only for one representative of each S-orbit on $L_{56}(6)$, and (ii) for one representative of each S-orbit on $L_{56}(8)$.

To collect representatives for $L_{56}(8)$ we use the larger group $S^* = S \cdot C$. If $x_1, ..., x_K$ represent the S^*-orbits on $L_{56}(8)$, then the vectors $x_i u$ (u in a set of 6 representatives of $C/\{\pm 1\}$) represent all S-orbits (with repetitions). The elements of C do *not* commute with τ, which is why all pairs (x_i, u) must be tested.

Table 3. The cases to be tested.

shell	vectors	S-orbits	structure and test units
$L_{56}(6)$ (minimal vectors)	15 590 400	656	632 orbits of size 24 360 and 24 of size 8 120; 656 representatives
$L_{56}(8)$	36 957 286 800	1 517 130	all of size 24 360; 254 664 S^*-orbits $\times$ 6 elements of $C/\{\pm 1\}$ = 1 527 984 pairs

4.3 Finite verification and the proof of Theorem A

Fix the data files of Appendix D: the Gram matrix G of L_{56}, the two generators of the symmetry group S, and the endomorphism matrix τ. The labels F0–F5 below identify the statements used by the verification programs; the headings describe their mathematical content.

Defining matrices (F0). The matrix G in Appendix D is positive definite, even integral and of determinant 1; (3.1) is also even and of determinant 1. [C2]

Minimum norm of the input lattice (F1). $\min L_{56} = 6$. [C5]

Complete norm-6 orbit list (F2). $L_{56}(6)$ is the disjoint union of the S-orbits of 656 explicitly listed vectors. [C8]

Complete norm-8 orbit list (F3). $L_{56}(8)$ is the union of the S-orbits of the 1 527 984 vectors $x_i u$; it consists of 1 517 130 regular S-orbits, that is, orbits with trivial stabilizer, hence of size $|S| =$ 24 360. [C9]

Symmetry and endomorphism identities (F4). The two generators of S preserve the Gram matrix, generate a group of order 24 360, and commute with τ; $\tau^2 = -5$ and $\tau G \tau^T = 5G$. [C2]

Lower bounds for the prescribed cosets (F5). (a) $\min(r\tau + 2L_{56}) \geq 14$ for each of the 656 representatives r. (b) $\min(x_i u\tau + 2L_{56}) \geq 12$ for each of the 1 527 984 pairs. Part (b) is established by two different methods (§§6.5 and 6.7); either one suffices. [C10, C11, C13]

Proof of Theorem A. The defining matrices give the even unimodular lattice of dimension 112 described in Lemma 3.2; the required endomorphism identities are part of F4. By F1 and Lemma 4.2, any nonzero vector of norm below 10 would give one of the four types listed in §1.3.

The isometries in S commute with τ by F4. Hence the minimum norm of the coset $(xg)\tau + 2L_{56}$ equals that of $x\tau + 2L_{56}$ for $g \in S$. By F2 and F3, the two lower bounds of Corollary 4.3 therefore need only be checked on the supplied norm-6 and norm-8 collections. F5 gives the bounds 14 and 12 on these collections. The type $(10, 6)$ is covered by the interchange argument of Appendix C.4. Thus no type can occur, and $\min L_{112} \geq 10$.

Since L_{112} is even unimodular of dimension 112, (1.1) gives $\min L_{112} \leq 10$. This proves the theorem. □

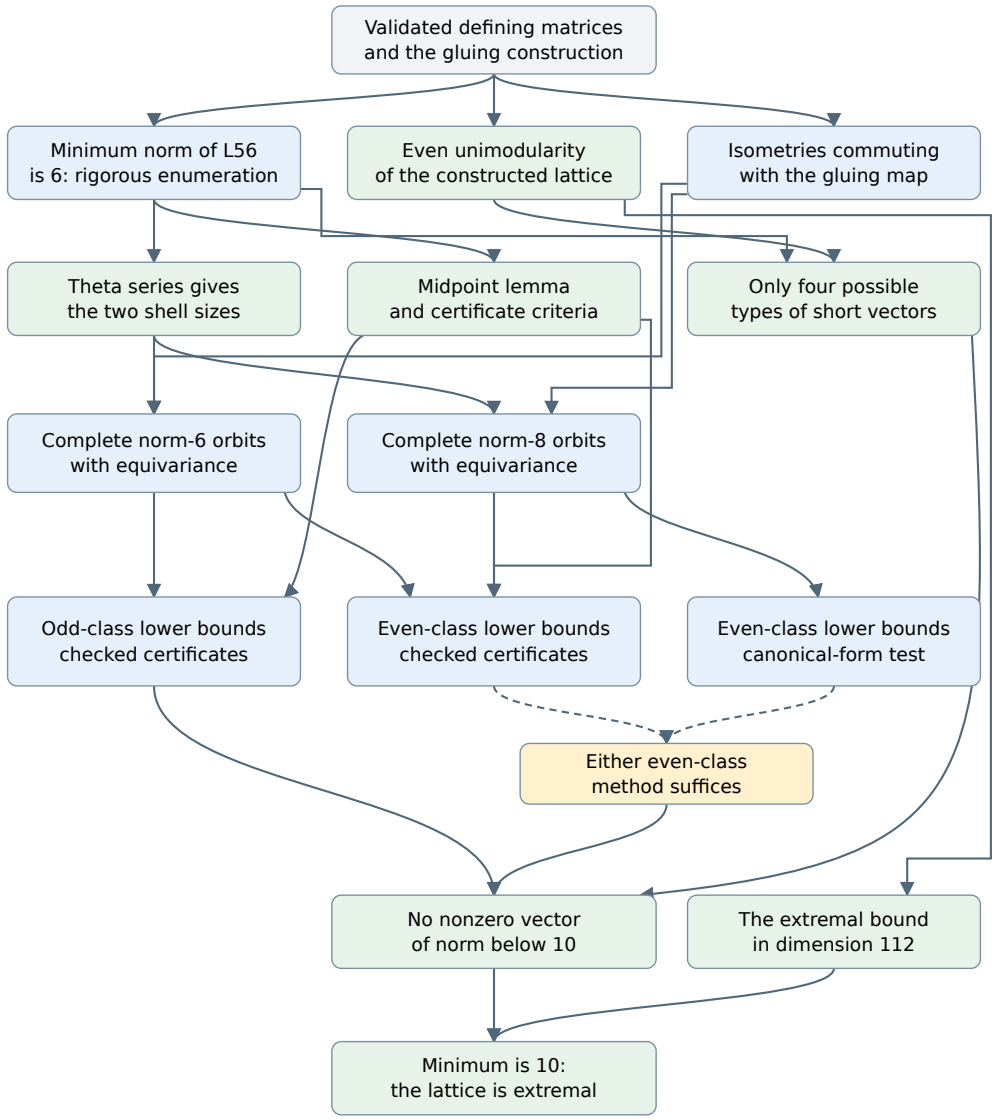


Figure 1. Dependencies in the proof of Theorem A. Solid arrows indicate results used in a deduction. The dashed arrows join two alternative even-coset tests: either one suffices, while the remaining joins use all incoming results. The orbit-completeness and equivariance checks are shared. The canonical-form box includes Proposition 6.3. The figure shows the main proof, not the order in which computations were run; supplementary arguments and consistency tests are described separately. Blue: finite computations. Green: mathematical deductions.

5. Certificates for coset minima

We derive lower bounds for a coset minimum from the norm of one representative and its inner products with all minimal vectors. The midpoint lemma proves that the stated conditions on a representative imply the required bound; the finite computations later check these conditions. A representative w_0 satisfying these conditions (Check 5.4) is called a **certificate** for its coset: checking it is a finite computation, and the corollaries below turn it into a proof of the bound.

Lemma 5.1 (midpoint lemma). Let L be even unimodular with $\min L = 6$. If $w, w' \in L$ lie in the same coset modulo $2L$ and $w \neq \pm w'$, then

$$\langle w, w\rangle + \langle w', w'\rangle \pm 2\langle w, w'\rangle \geq 24 \qquad \text{for both signs.}$$

Proof. $w \mp w' \in 2L \setminus \{0\}$, so the midpoint $\frac{1}{2}(w \mp w')$ is a nonzero vector of L and has norm ≥ 6; multiply by 4. □

The two corollaries below retain the assumptions on L in Lemma 5.1. Two immediate consequences follow. Two vectors of norm 6 that are distinct up to sign never lie in the same coset (with $t = \langle w, w'\rangle$, the inequalities $12 + 2t \geq 24$ and $12 - 2t \geq 24$ cannot both hold), and the same holds for norm 8. Hence each coset contains at most one pair $\pm w$ of norm 6, and at most one pair of norm 8 (Appendix C.3).

Corollary 5.2 (lower bound for odd cosets). If an odd coset c contains a vector w_0 of norm $N \in \{14, 18\}$ such that $|\langle w_0, u\rangle| \leq 6$ for every $u \in L(6)$, then $\min(c) \geq 14$.

Proof. Suppose $w \in c$ has norm $m \leq 10$. Norms in c are $\equiv N \pmod 4$, so $m \in \{6, 10\}$; in particular $m < N$, $N + m \leq 28 < 32$ and $w \neq \pm w_0$. The midpoints $a = \frac{1}{2}(w_0 + w)$ and $b = \frac{1}{2}(w_0 - w)$ are nonzero vectors of L (Lemma 5.1) with $\langle a, a\rangle + \langle b, b\rangle = \frac{1}{2}(N + m) < 16$. Since L is even with minimum 6, they cannot both have norm ≥ 8; so one of them is a minimal vector u, and $w_0 - 2u = \pm w$. Taking norms, $N - 4\langle w_0, u\rangle + 24 = m$, i.e. $\langle w_0, u\rangle = \frac{1}{4}(N + 24 - m) > 6$, a contradiction. □

Corollary 5.3 (lower bound for even cosets). If an even coset c contains a vector w_0 of norm $N \in \{12, 16, 20\}$ such that $|\langle w_0, u\rangle| \leq 6$ for every $u \in L(6)$, then $\min(c) \geq 12$.

Proof. Norms in the even coset c are $\equiv 0 \pmod 4$. The coset does not contain 0 (otherwise $w_0 \in 2L$ and $N \geq 4 \cdot 6$), nor a vector of norm 4 (since $\min L = 6$). If $w \in c$ had norm $m = 8$, then $m < N$ and $N + m \leq 28$, and the argument of Corollary 5.2 gives a minimal vector u with $\langle w_0, u\rangle = \frac{1}{4}(N + 16) > 6$, a contradiction. □

Check 5.4 (coset-bound data). To verify a lower bound for a coset c, supply a representative $w_0 \in c$ satisfying Corollary 5.2 or 5.3. Check:
(a) $w_0 \in c$; when $c = x\tau + 2L$, every coordinate of $w_0 - x\tau$ is even.
(b) $\langle w_0, w_0\rangle$ belongs to $\{14, 18\}$ for an odd coset, or to $\{12, 16, 20\}$ for an even coset.
(c) $|\langle w_0, u\rangle| \leq 6$ for every $u \in L(6)$, using the complete minimal-vector list.

These conditions are sufficient but not necessary: a coset without short vectors need not have a representative satisfying them. For every coset required in §§6.6–6.7, such a representative is supplied and checked.

6. Algorithms and the verification

Three implementations are referred to below: implementation 1 constructed L_{112} and performed the main computations; implementation 2 re-implemented every step, sharing only the data files; implementation 3 started from the four data files of Appendix D alone (Supplement §S3.3). For each computational fact we describe the algorithm and the checks on its output. Completeness requires both disjointness and agreement of the orbit-size sum with the theta-series coefficient. The two rigorous even-coset methods share these orbit lists and the odd-coset bounds (§1.3); separate implementations and tests on gluings A and B check their results. Floating-point searches and samples have the supplementary scope stated below. Running times are those observed on the machine used (8 cores, 15 GB of memory). Code references are given in Appendix D and pseudocode in Appendix E.

Table 4. The computational facts and how they were checked.

fact	main method	other methods, independent reproductions	nature
F1	interval-arithmetic enumeration (BKZ-40 basis)	second exact enumeration in integer arithmetic (a′); reduction to $L_{56,2}(M)$ plus $\mathbb{F}_2$ linear algebra; fplll (two bases); PARI `qfminim`	interval arithmetic, integer (others: floating point)
F2	closure + orbits, size sum	disjointness, design identities (implementation 2); implementation 3	integer
F3	marked sums of two minimal vectors, size sum	recount from K_8, design identities; implementation 3	integer
F4	matrix identities	two implementations	integer
F5a	656 odd-coset certificates	the checker of implementation 2; CVP on two bases; canonical forms	integer
F5b	canonical forms (two implementations)	exhaustive CVP; implementation 3; even-coset certificates (all pairs); support-module route (Supplement §S4.1)	integer (CVP: floating point)

6.1 F1: the minimum of the base lattice

An interval-arithmetic enumeration, which cannot miss vectors, shows that L_{56} has no nonzero vector of norm ≤ 4. As an independent second route, the question is reduced to a property of Nebe's lattice $L_{56,2}(M)$ together with linear algebra over $\mathbb{F}_2$. Both agree with the three earlier floating-point enumerations.

(a) Interval enumeration. [C5]

1. Reduce the Gram matrix with BKZ-40. Check exactly that the transformation U is an integer matrix with $\det U = \pm 1$ and compute $G' = UGU^T$ in integers (floating point used inside the reduction does not affect correctness).
2. Compute $G' = V^T \operatorname{diag}(D)V$ (V unit upper triangular) **exactly in rational arithmetic**.
3. Replace every entry of D and V by an interval around its nearest double, with a safety margin on both sides.

4. Run Fincke–Pohst/Schnorr–Euchner enumeration [FP85, SE94] in interval arithmetic, widening every result outward (by 2^{-51} relative, against the 2^{-53} error of round-to-nearest), and prune only when the **lower end** of the interval of the partial norm exceeds the radius 4.

The true partial norm always lies in its interval, so no vector of true norm ≤ 4 can be pruned. The norms of the vectors reaching the leaves are computed in integers. The results are in Table 5: for L_{56}, 8 454 837 756 nodes were visited and the only leaf reached was the zero vector (34 minutes with the published code).

Table 5. Results of the interval enumeration (only one of $\pm x$ is enumerated; counts include both signs).

lattice	dim	radius	nodes	nonzero vectors found	expected
E_8 (control)	8	2	248	240 of norm 2	240
Leech lattice (control)	24	4	2 019 561	196 560 of norm 4, none of norm 2	same
$\Lambda_{24} \perp \Lambda_{24} \perp E_8$ (control)	56	4	11 081 124	240 of norm 2, 395 280 of norm 4	same
$\boldsymbol{L_{56}}$	56	4	8 454 837 756	**none**	—

(a′) A second exact enumeration, by a different method. A separately written program (developed without reference to the code of (a)) makes every decision in integer arithmetic instead of interval arithmetic. The approximate Cholesky factor of G' (BKZ-40 basis, U checked exactly as in (a)) is scaled by 2^{40} and rounded to an integer upper triangular matrix R; then $Q(x) = |Rx^T|^2$ is an integer quadratic form, the error matrix $E = 2^{80}G' - R^TR$ is computed exactly, and $|xEx^T| \leq \|E\|_F \left\|G'^{-1}\right\|_\infty \langle x, x\rangle$ with G'^{-1} an exact integer matrix. Hence $\langle x, x\rangle \leq 4$ implies $Q(x) \leq T := 4 \cdot 2^{80} + 4\|E\|_F \left\|G'^{-1}\right\|_\infty$ (a relative enlargement of 3.8×10^{-7}). All x with $Q(x) \leq T$ are enumerated in C with 128-bit integers, the range at each level being given by exact integer square roots, so no vector is pruned because of rounding. For L_{56}: 14 773 398 896 nodes (both signs), no nonzero vector found (96 minutes) [C27]. Positive control: E_8^7 in a random basis gives 1 680 vectors of norm 2 and 1 224 720 of norm 4, as expected.

(b) Reduction to Nebe's lattice. L_{56} is the sublattice $L_{56,2}(M)X + 2L_{56,2}(M)$ of the catalog lattice (where X is an element of $M/2M$ whose action modulo 2 has an image of dimension 28), rescaled by $1/\sqrt{2}$; the Gram matrix of this sublattice agrees entry by entry with that of L_{56}. Hence norm-4 vectors of L_{56} correspond bijectively to norm-8 vectors of $L_{56,2}(M)$ whose class lies in an $\mathrm{SL}_2(29)$-invariant 28-dimensional subspace W of $L_{56,2}(M)/2L_{56,2}(M)$, and norm-2 vectors of L_{56} to norm-4 vectors of $L_{56,2}(M)$. Consider the 1 528 956 pairs (norm-8 orbit representative of $L_{56,2}(M)$, centralizer element); the 127 413 representatives have orbit sizes summing to the theta coefficient. None of these pairs has its coset in W [C6]. Positive control: the 656 orbit representatives of minimal vectors of L_{56} all appear in $L_{56,2}(M)$ with norm 12 and class in W. This proves "$\min L_{56,2}(M) = 6 \Rightarrow \min L_{56} = 6$", reducing F1 to a property of the literature lattice [Neb98, Table II].

(c) Floating-point enumerations. fplll enumeration of all vectors of norm ≤ 4.5 (two bases) and PARI `qfminim` for norm ≤ 4 (implementation 2, BKZ-50 basis) also found none [C7].

6.2 F2: orbits of minimal vectors

We grow the set of minimal vectors by a closure operation, split it into orbits, and stop when the orbit sizes add up to the kissing number 15 590 400. Disjointness and agreement of the total sizes prove completeness.

Starting from a few minimal vectors found by BKZ, we generated minimal vectors by closure: if $r, s \in L_{56}(6)$ and $\langle r, s\rangle = 3$ then $r - s \in L_{56}(6)$. For each new vector we computed its S-orbit (by applying all 12 180 elements of $S/\{\pm 1\}$) and recorded it as one orbit; the closure step produces vectors outside

the orbits already found. We obtained 656 distinct orbits, with sizes adding up to $632 \times 24\,360 + 24 \times 8\,120 = 15\,590\,400 = |L_{56}(6)|$ [C8]. Since distinct orbits are disjoint, **completeness follows from the size sum and F1**. A separate implementation expanded all orbits, checked pairwise disjointness, and checked the identities of §6.3.

6.3 Completeness tests: orbit-size sums and design identities

In addition to the size sum, we verified exact identities (equalities of integers of size about 10^{18}) that follow from the shells being spherical 7-designs, computed from the lists of representatives. Barring coincidence, duplicates or omissions would violate them.

For every $a \in \mathbb{R}^{56}$

$$\sum_{x \in L_{56}(m)} \langle x, a\rangle^2 = |L_{56}(m)| \frac{m\langle a, a\rangle}{56}, \quad \sum_{x \in L_{56}(m)} \langle x, a\rangle^4 = |L_{56}(m)| \frac{3m^2\langle a, a\rangle^2}{56 \cdot 58}, \quad \sum_{x \in L_{56}(m)} \langle x, a\rangle^6 = |L_{56}(m)| \frac{15m^3\langle a, a\rangle^3}{56 \cdot 58 \cdot 60}.$$

The left sides were computed from the representative lists (representative × group element × element of C, weighted by orbit sizes) for three integer vectors a and for $m = 6$ and $m = 8$; all 18 identities hold exactly [C8].

6.4 F3: orbits of norm-8 vectors via marked sums of two minimal vectors

The approximately 37 billion vectors are too many to store, so we distinguish orbits by marking only those of their elements that are sums of two minimal vectors. Completeness is again proved by the size sum.

Every norm-8 vector we need can be written as $x = r + s$ with $r, s \in L_{56}(6)$ and $\langle r, s\rangle = -2$, and by symmetry r may be taken among the (at most 150) S^*-orbit representatives r_i of minimal vectors. Let $X_i = \{r_i + s : s \in L_{56}(6), \langle r_i, s\rangle = -2\}$.

> **Definition 6.1.** For an S^*-orbit $O \subset L_{56}(8)$, the **marked elements** of O are the elements of $O \cap \bigcup_i X_i$, that is, the elements of O that are sums $r_i + s$ of the above shape (possibly in several ways). Completeness of the orbit list does not rely on every norm-8 vector being such a sum: it follows from the orbit-size sum (Table 3).

The algorithm runs through candidates $x \in X_i$; whenever x is not yet marked, it starts a new orbit, computes all elements of that orbit in $\bigcup_i X_i$ and marks them. This needs only one matrix product per orbit: $y = \pm xug^{-1}$ lies in X_i iff $\langle y, r_i\rangle = 4$, i.e. iff $\langle xu, r_i g\rangle = \pm 4$, so precomputing $W = \{r_i g\}$ finds them all at once. The decompositions of $x = r_i + s_0$ can be read off from the minimal vectors with inner product 2 or 3 with r_i, which reduces the work by a factor of about 30. The stabilizer order of x is the number of elements of S^* that map x to itself, read off from the same matrix product (they are group elements, not decompositions); a wrong count would make the orbit sizes, and hence their sum, wrong. Only the marked elements (tens of millions) are stored.

The result is Table 3: $1\,517\,130 \times 24\,360 = 36\,957\,286\,800 = |L_{56}(8)|$ [C9]. This was checked in three ways: (1) the S^*-orbit sizes add up to $|L_{56}(8)|$; (2) the set K_8 of canonical forms (§6.5) of the cosets of all $x_i u$ has exactly $1\,517\,130$ elements, and recomputing each orbit size from K_8 gives the same total (which would fail if an orbit were counted twice); (3) the identities of §6.3. The first 25 orbits were cross-checked against a naive method, and the canonical forms of the cosets of 2 000 independently generated norm-8 vectors were all found in K_8.

6.5 F5b: canonical forms on $L_{56}/2L_{56}$

A coset is stored as a 56-bit integer and its $\mathrm{SL}_2(29)$-orbit is named by the smallest element (canonical form). Since a class contains a norm-8 vector if and only if its canonical form lies in the table K_8, each test is a single table look-up. Only integer arithmetic is used, so no rounding occurs.

Identify $L_{56}/2L_{56}$ with $\mathbb{F}_2^{56}$ via coordinates modulo 2. Each $g \in S/\{\pm 1\}$ acts by an $\mathbb{F}_2$-linear map, implemented with byte-wise lookup tables and XOR.

Definition 6.2 (canonical form). The **canonical form** $\mathrm{can}(c)$ of a coset c is the smallest 56-bit integer in its orbit $\{cg : g \in S\}$.

Proposition 6.3. Let $K_8 = \{\mathrm{can}(x_i u + 2L_{56})\}$ (all pairs). Assuming F3, a coset c contains a vector of norm 8 if and only if $\mathrm{can}(c) \in K_8$.

Proof. If $w \in c$ has norm 8, by F3 $w = x_i u g$ for some i, u and $g \in S$, so $c = (x_i u + 2L_{56})g$ and $\mathrm{can}(c) = \mathrm{can}(x_i u + 2L_{56}) \in K_8$. Conversely, $\mathrm{can}(c) = \mathrm{can}(x_i u + 2L_{56})$ means $c = (x_i u + 2L_{56})g$ for some g, and $x_i u g \in c$ has norm 8. □

Test for type $(8,8)$: for each pair, compute the coset of $x_i u \tau$, its canonical form (12 180 table look-ups, about 0.3 ms), and look it up in K_8. The results are in Table 6. The test was implemented twice with separately written code (group enumeration, tables, canonical forms, look-up), with self-tests (table look-up equals matrix product; canonical form is S-invariant; the 12 180 group elements are distinct modulo 2). The type $(6,6)$ was tested on the 656 minimal-vector cosets using $K_6 = \{\mathrm{can}(r_i + 2L_{56})\}$.

Table 6. Results of the canonical-form test for type $(8,8)$ (identical for τ and $\tau^{-1} \bmod 2$).

gluing	pairs tested	colliding pairs	colliding S-orbits	norm-8 vectors of L_f
class containing an element of C	3 000 (sample)	3 000	—	— (minimum 6)
gluing A	1 528 956	36	6	292 320
gluing B	1 527 984	10	6	292 320
gluing C	1 527 984	**0**	**0**	**0**

6.6 F5a: certificates for the types $(6,6)$, $(6,10)$, $(10,6)$

All 656 odd cosets have a certificate of norm 14 or 18. The search is heuristic; its output is checked by Check 5.4.

For each representative r we start from $w = r\tau$ (norm 30) and, while some minimal vector u has $\langle w, u\rangle \geq 7$, replace w by $w - 2u$ (the **descent**). This lowers the norm by $4(\langle w, u\rangle - 6) > 0$ and stays in the class. Positive control: starting from the sum of a norm-10 vector and an element of $2L_{56}$, the descent ended at norm 10 in all 60 trials and never produced a false certificate.

Table 7. The 656 odd-coset certificates.

$\langle w_0, w_0\rangle$	number	maximum of $\lvert\langle w_0, u\rangle\rvert$ over all minimal vectors
14	243	5 (43 vectors), 6 (200)
18	413	6 (all 413)

The certificates were produced by one implementation and re-checked by a separate implementation with separately written code, which also re-derived the lemma [C10]. Two closest-vector enumerations on different bases independently found no vector of norm ≤ 10 in these classes [C12].

6.7 F5b: bounds from even-coset representatives

Even-coset certificates. For every pair we computed an even-coset certificate (Corollary 5.3) by the descent of §6.6, starting from $w = x_i u\tau$ (norm 40). Inner products are computed in float64; they are exact because every partial sum of a product AB of integer matrices is bounded by $\sum_k |a_k|\,|b_k| \le \left(\max_{\text{row}} \sum_k |a_k|\right) \cdot \max|B|$, which is checked to be below 2^{52} at run time for both products used. The result: **every one of the 1 527 984 pairs has a certificate** (Table 8) [C13]. Thus F5b also holds by an argument that uses only the midpoint lemma, integer inner products and the list F2, independently of canonical forms. All certificates were re-checked by a separately written program, using integer arithmetic and float32 products only where all partial sums are proved to be below 2^{24} [C27].

Table 8. Even-coset certificates for all 1 527 984 pairs.

$\langle w_0, w_0\rangle$	max $\lvert\langle w_0, u\rangle\rvert$	number	share
12	5	41 437	2.71 %
12	6	692	0.05 %
16	6	1 465 524	95.91 %
20	6	20 331	1.33 %

7. Identification and formalization

7.1 Formalization of the deduction (Lean)

With the computational facts as **hypotheses**, the conclusion "L_{112} has no inner-shell vectors" has been proved in the proof assistant Lean 4 (with mathlib) and machine-checked. This excludes errors that consist of a gap in the argument (a missing type, a missing case, or a wrong direction).

The setting is abstract: L_{56} is an additive group with an integer-valued symmetric bilinear form B, even norms, nonzero vectors of norm ≥ 6 (F1), and τ with $B(x\tau, y\tau) = 5B(x, y)$ and $\tau^2 = -5$. The formalized statements are listed in Table 9.

Table 9. Statements proved in Lean (`lean/EEUL/L112/Chain.lean`).

Lean theorem	in this paper	content
`no_inner_shell`	Lemma 4.2, Corollary 4.3	(i) $\min(r\tau + 2L_{56}) \ge 14 \quad (r \in L_{56}(6))$ and (ii) $\min(x\tau + 2L_{56}) \ge 12 \quad (x \in L_{56}(8))$ imply $\langle x, x\rangle + \langle x\tau + 2y, x\tau + 2y\rangle \ge 20$ whenever $(x, x\tau + 2y) \ne 0$
`odd_bound`, `even_bound`	Lemma 5.1, Corollaries 5.2, 5.3	a certificate forces all vectors of its coset to have norm ≥ 14 (odd) or ≥ 12 (even)
`theoremA`	§4.2 and the derivation of Theorem A	assuming F1 and F2–F5 (the symmetry group is isometric and commutes with τ; norm-6 and norm-8 vectors are images of representatives; each representative's class has a certificate), there are no inner-shell vectors

Check: the package builds, contains no forbidden keywords such as `sorry`, and each theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound` [C22]. Not formalized: unimodularity (an integer determinant computation) and the computational facts themselves (the individual numbers in F1–F5).

7.2 Identification with Nebe's lattice

Our L_{56} is $\mathrm{SL}_2(29)$-equivariantly isometric to $L_{56,2}(\widetilde{M})$ built from Nebe's definition [Neb98] without external data [C23].

1. Build $L_{56'} = L_{56,2}(\widetilde{M})$ and generators z' (order 29), w' (order 4) of $\mathrm{SL}_2(29)$ from the model over $\mathbb{Q}(\zeta_{29})$ (code: `code/supplementary/nebe/cycloquat.py` of the two classes of maximal orders, the one whose centralizer has order 12 is $\widetilde{M}$).
2. Find a group isomorphism between the two copies of $\mathrm{SL}_2(29)$ with GAP (also composed with the outer automorphism), and compute the space of equivariant maps $L_{56'} \to L_{56}$; it is 8-dimensional.
3. On its integral part (rank 8) enumerate the elements with $Q(T) = \mathrm{tr}(G'^{-1}TGT^T) \le 56$; isometries are exactly the elements with $Q = 56$.

We find 6 (up to sign) unimodular integer matrices T with $TGT^T = G'$ intertwining the two actions, as many as predicted by the centralizer of order 12. Moreover, in the coordinates of $L_{56'}$, τ becomes right multiplication by an element of $\widetilde{M}$ of reduced norm 5. Thus **gluing C is built from Nebe's lattice $L_{56,2}(\widetilde{M})$ and an element of reduced norm 5 and reduced trace 0 of its maximal order $\widetilde{M}$; such elements fall into two classes modulo 2 in the sense of §3.3, and τ belongs to the class giving minimum 10**.

The element explicitly. `cycloquat.py` works in the algebra $\mathbb{Q}(\zeta_{29}) \oplus \mathbb{Q}(\zeta_{29})b$ with $b^2 = -1$ and $bx = \bar{x}b$, where $\bar{x}$ is the complex conjugate; $\mathrm{SL}_2(29)$ acts on it from the left by the construction of [Neb98, §3]. Let $K_4 \subset \mathbb{Q}(\zeta_{29})$ be the quartic subfield; its ring of integers has the $\mathbb{Z}$-basis of Gaussian periods $\eta_j = \sum_{k=0}^{6} \zeta_{29}^{2^{4k+j}}$ ($j = 0, \dots, 3$; 2 is a primitive root modulo 29). Complex conjugation exchanges $\eta_0 \leftrightarrow \eta_2$ and $\eta_1 \leftrightarrow \eta_3$, and its fixed field in K_4 is $\mathbb{Q}\big(\sqrt{29}\big)$. The commutant of $\mathrm{SL}_2(29)$ is right multiplication by $\mathcal{Q} = K_4 \oplus K_4 b$, the quaternion algebra of §2.4. The maximal order $\widetilde{M}$ constructed by the program is the $\mathbb{Z}$-module with basis

$$m_0 = \frac{\eta_0 + 41\eta_1 + 144\eta_2 + 104\eta_3 + (133\eta_0 + 88\eta_1 + 12\eta_2 + 492\eta_3)b}{145},$$

$$m_1 = \eta_1 + 2\eta_3 b, \quad m_2 = \eta_2 + 3\eta_3 b, \quad m_3 = \eta_3 + 3\eta_3 b,$$

$$m_4 = (\eta_0 + \eta_3)b, \quad m_5 = (\eta_1 + \eta_3)b, \quad m_6 = (\eta_2 + 4\eta_3)b, \quad m_7 = 5\eta_3 b.$$

In the coordinates of $L_{56'}$, the endomorphism τ is right multiplication by

$$h = x + yb, \quad x = \frac{33(\eta_0 - \eta_2) + 48(\eta_1 - \eta_3)}{145}, \quad y = \frac{-106\eta_0 + 4\eta_1 - 39\eta_2 - 4\eta_3}{145},$$

that is, $h = 33m_0 - 9m_1 - 33m_2 - 24m_3 - 31m_4 - 20m_5 - 3m_6 + 28m_7 \in \widetilde{M}$. Since $\bar{x} = -x$, the reduced trace of h is 0, and its reduced norm $x\bar{x} + y\bar{y}$ is 5, so that $h^2 = -5$. Replacing T by another of the 6 isometries replaces h by $u^{-1}hu$ for a unit u of $\widetilde{M}$, which does not change the isometry class of the gluing (Proposition 3.3(b)). The lattice $L_{56,2}\big(\widetilde{M}\big)$ is the $\mathrm{SL}_2(29)$-invariant unimodular lattice among the lattices $(1-\zeta_{29})^i\,\mathbb{Z}[\zeta_{29}]\widetilde{M}$ ($i \ge 0$; spanned by the products of elements of $\mathbb{Z}[\zeta_{29}]$ on the left and of $\widetilde{M}$ on the right) with the forms $\mathrm{Tr}_{\mathbb{Q}(\zeta_{29})/\mathbb{Q}}\big(c(x\bar{x'} + y\bar{y'})\big)$ on elements $x + yb$, $x' + y'b$, where $c \in \mathbb{Q}\big(\sqrt{29}\big)$ is totally positive [Neb98, §3]. Together with (1.2), these data determine L_{112} up to isometry without the files.

8. The automorphism group

We first compute, by finite exact lattice computations, the normalizer of $\mathrm{SL}_2(29)$ in $\mathrm{Aut}(L_{112})$; then we prove that $\mathrm{SL}_2(29)$ is normal in $\mathrm{Aut}(L_{112})$, by a group-theoretic argument modeled on Sakamoto's determination of the automorphism groups in dimension 88 [Sak26, App. A] (Appendix F). Neither step uses the extremality of L_{112}, and Theorem A does not depend on this section.

Theorem B. Let $S \cong \mathrm{SL}_2(29)$ act diagonally on L_{112}. (a) The centralizer $C_{\mathrm{Aut}(L_{112})}(S)$ has order exactly 48 [C24]. (b) The outer automorphism of S is induced by an automorphism of L_{112}; hence $|N_{\mathrm{Aut}(L_{112})}(S)| = 2 \cdot 24\,360 \cdot 48/2 = 1\,169\,280$ (the factor $1/2$ because S and its centralizer share -1, the factor 2 for the outer automorphism) [C24]. (c) S is the layer $E(\mathrm{Aut}(L_{112}))$ (the product of the components), in particular normal, so $\mathbf{|Aut}(\boldsymbol{L_{112}})\mathbf{| = 1\,169\,280}$ [C25] (Appendix F; it uses the classification of finite simple groups through the tables of Hiss–Malle [HM01, HM02] and the Schreier conjecture).

Let G_{112} be the Gram matrix (3.1) of L_{112}.

Algorithm for (a) and (b). An endomorphism of $V = L_{112} \otimes \mathbb{Q}$ commuting with S is a 2×2 block matrix $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $a, b, c, d \in M' \otimes \mathbb{Q}$ ($M' = \mathrm{End}_S(L_{56})$, 8-dimensional), so these form a 32-dimensional space. Those mapping L_{112} into itself form a $\mathbb{Z}$-lattice $\mathcal{H}$ of rank 32, obtained exactly as the dual of the $\mathbb{Z}$-span of the coefficient rows. On $\mathcal{H}$ the form $Q(X) = \mathrm{tr}(G_{112}^{-1} X G_{112} X^T)$ is positive definite and equals 112 on every isometry, so all isometries are among the finitely many $X \in \mathcal{H}$ with $Q(X) \le 112$, which we enumerate (Fincke–Pohst, PARI) and test exactly. Since PARI's enumeration is not documented to be complete, the same vectors are enumerated again with the interval-arithmetic enumerator of §6.1 (a), which cannot miss vectors, and the two lists agree [C24]. For (b) we take the outer automorphism α of S from GAP and repeat this with the 8-dimensional space of maps T on L_{56} with $gT = T\alpha(g)$.

Table 10. Determination of $\mathrm{Aut}(L_{112})$.

Step	Result	Method	Time
(a) centralizer	$Q \le 112$: 24 elements up to sign, all isometries; $\lvert C_{\mathrm{Aut}(L_{112})}(S)\rvert = 48$	rank-32 lattice $\mathcal{H}$, enumeration	7 s
(b) outer automorphism	twisted intertwiners: 8-dimensional on L_{56}; 24 isometries up to sign	GAP for α, then as in (a)	11 min
diagonal case (Appendix F.4)	none of the 6 involutions $\neq \pm 1$ of $C_{\mathrm{Aut}(L_{112})}(S)$ splits S	integrality test	10 s
(c) normality	$E(\mathrm{Aut}(L_{112})) = S$	Appendix F (proof)	—

The centralizer of order 48 is the unitary automorphism group of the rank-2 M'-lattice $\Lambda = \frac{\{(a,b)\in (M')^2 : b \equiv a\tau \pmod{2M'}\}}{\sqrt{2}}$, with $L_{112} = L_{56} \otimes_{M'} \Lambda$. The subgroup $S \cdot C_{\mathrm{Aut}(L_{112})}(S)$ has order 584 640; the outer automorphism doubles this order. A matrix inducing the outer automorphism is supplied as `certificates/outer_automorphism_R.txt` [C24, C25].

Structure. From explicit generators (those of S, the 48 elements of $C_{\mathrm{Aut}(L_{112})}(S)$, and one element inducing the outer automorphism), reduced modulo 3 (which is injective on a finite group of integer matrices), GAP finds a group of order exactly 1 169 280 [C26], an independent check of the order. Moreover $C_{\mathrm{Aut}(L_{112})}(S) \cong \mathrm{SL}_2(3) \circ C_4$ (SmallGroup(48,33)), $\mathrm{Aut}(L_{112})/S \cong C_2 \times S_4$,

$\mathrm{Aut}(L_{112})/C_{\mathrm{Aut}(L_{112})}(S) \cong \mathrm{PGL}_2(29)$, $Z(\mathrm{Aut}(L_{112})) = \{\pm 1\}$, and the extension of the index-2 subgroup $S \cdot C_{\mathrm{Aut}(L_{112})}(S)$ does not split (the outer coset contains no involution) [C26]. In short, $\mathrm{Aut}(L_{112}) \cong (\mathrm{SL}_2(29) \circ \mathrm{SL}_2(3) \circ C_4).2$, central products over the common -1. All generators have determinant 1, and the endomorphisms of L_{112} commuting with $\mathrm{Aut}(L_{112})$ form the ring $\mathbb{Z}\left[\sqrt{-29}\right]$; so L_{112} is a Hermitian $\mathbb{Z}\left[\sqrt{-29}\right]$-lattice of rank 56 and $\mathrm{Aut}(L_{112}) \subset \mathrm{U}(56) \subset \mathrm{SO}(112)$.

9. Authorship and use of AI

Direction, choice of problem, decisions, and responsibility for publication: roiban1344. Search strategy, code, computations, verification, and the initial draft: Claude (Anthropic), in several sessions that worked in parallel and checked each other's work with separately written code. The framework of gluing two lattices modulo 2 and reducing extremality to four collision types was first worked out, in dimension 96, by a session of OpenAI Codex. OpenAI Codex was used to restructure and revise the paper and supplement in response to the review. The author takes responsibility for the claims and for any errors.

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[**Repo**] roiban1344, *112-dimensional extremal even unimodular lattice* (data, the code that checks and produces the verification data, the verification data, the Lean formalization), public repository (tag `v2026-10-18.1`).

Appendix A. Glossary

Coset; coset minimum min(c) An element $c = x + 2L$ of $L/2L$, and the smallest norm of a vector in it. Even/odd according to $\langle x, x\rangle/2 \bmod 2$ (§2.2).
Gluing L_f The lattice $\left\{\frac{1}{\sqrt{2}}(x, v) : v \equiv xf\right\}$ built from two copies of L and an invertible parity-preserving f (Definition 3.1). Our L_{112} is L_f with $f = \tau \bmod 2$.
Collision of type (a, b) A pair (x, v) with x of norm a and v of norm b in the coset $x\tau + 2L$ (Definition 4.1). The inner-shell vectors of L_{112} are exactly the collisions of type $(6, 6), (6, 10), (10, 6), (8, 8)$ (Lemma 4.2).
Certificate (for a coset) A vector w_0 of a coset c with norm 14 or 18 (odd coset) or 12, 16, 20 (even coset) and $|\langle w_0, u\rangle| \le 6$ for all minimal vectors u; it guarantees $\min(c) \ge 14$ (odd) or $\min(c) \ge 12$ (even) (Corollaries 5.2, 5.3; Check 5.4).
Canonical form; K_8 $\mathrm{can}(c)$ is the smallest 56-bit integer in the $\mathrm{SL}_2(29)$-orbit of a coset c; K_8 is the set of canonical forms of cosets of norm-8 vectors. A coset contains a norm-8 vector iff its canonical form is in K_8 (Proposition 6.3).
Orbit representative A vector r_i or $x_i u$ selected from an S-orbit; not to be confused with the coset representative w_0 that certifies a bound (§§4.2, 5).

Appendix B. Extremal theta series

Scope. This appendix gives the sources of the numbers used in the text; all are exact rational computations. With $A = \theta_{E_8} = E_4$ and $B = \theta_{\Lambda_{24}} = E_4^3 - 720\Delta$, by [Ser73, Ch. VII], every modular form of weight $4k$ is a polynomial in A and B, and the extremal theta series in dimension $n = 8k$ is the unique one of the form $1 + O(q^{\lfloor n/24\rfloor+1})$.

Table B1. Extremal theta series in dimensions 56 and 112.

n	as a polynomial in A, B	first coefficients
56	$\frac{1}{240}(-159A^7 + 238A^4B + 161AB^2)$	$1 + 15\,590\,400q^3 + 36\,957\,286\,800q^4 + 15\,284\,192\,071\,680q^5 + \cdots$
112	$\frac{1}{37\,324\,800}(7\,883\,087A^{14} - 18\,165\,308A^{11}B - 38\,110\,758A^8B^2 + 74\,301\,892A^5B^3 + 11\,415\,887A^2B^4)$	$1 + 10\,888\,335\,360q^5 + 247\,006\,775\,232\,000q^6 + \cdots$

Appendix C. Proofs of auxiliary facts

C.1 τ is skew-adjoint and h is Hermitian

From $\tau G \tau^T = 5G$ and $\tau^2 = -5$ we get $G\tau^T = 5\tau^{-1}G = -\tau G$, i.e. $\langle x\tau, y\rangle = -\langle x, y\tau\rangle$. With $h(x,y) = \langle x, y\rangle + \langle x\tau, y\rangle/\sqrt{-5}$: $h(x\tau, y) = \langle x\tau, y\rangle + \langle x\tau^2, y\rangle/\sqrt{-5} = \langle x\tau, y\rangle + \sqrt{-5}\langle x, y\rangle = \sqrt{-5}\, h(x,y)$, so h is $\mathcal{O}$-linear in the first variable; and $h(y,x) = \langle x, y\rangle - \langle x\tau, y\rangle/\sqrt{-5} = \overline{h(x,y)}$ because $\overline{1/\sqrt{-5}} = -1/\sqrt{-5}$. Clearly $\mathrm{Re}\, h = \langle \cdot, \cdot \rangle$.

C.2 Proof of Proposition 3.4

Since $H = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ is free, every element of $H \otimes_{\mathcal{O}} P$ is uniquely $e_1 \otimes x + e_2 \otimes y$, and φ is a bijection onto L_τ (put $v = x\tau + 2y$). For the forms: $\mathrm{Re}(h_H(e_1,e_1)h_P(x,x')) = 3\langle x, x'\rangle = \frac{1}{2}(\langle x, x'\rangle + \langle x\tau, x'\tau\rangle)$; $\mathrm{Re}(h_H(e_2,e_2)h_P(y,y')) = 2\langle y, y'\rangle = \frac{1}{2}\langle 2y, 2y'\rangle$; and $\mathrm{Re}(\sqrt{-5}\, h_P(x,y')) = \mathrm{Re}(\sqrt{-5}\langle x, y'\rangle + \langle x\tau, y'\rangle) = \langle x\tau, y'\rangle = \frac{1}{2}\langle x\tau, 2y'\rangle$. These are the inner products of $\varphi(e_1 \otimes x) = (x, x\tau)/\sqrt{2}$ and $\varphi(e_2 \otimes y) = (0, 2y)/\sqrt{2}$, i.e. the blocks of (3.1).

C.3 Injectivity of the coset map on short shells

If $w \neq \pm w'$ have norm $m \in \{6, 8\}$ and lie in the same coset, the midpoint lemma gives $2m + 2\langle w, w'\rangle \geq 24$ and $2m - 2\langle w, w'\rangle \geq 24$; adding them gives $4m \geq 48$, which is false. So $L(6)/\{\pm 1\} \to L/2L$ and $L(8)/\{\pm 1\} \to L/2L$ are injective, with disjoint images (odd and even cosets respectively). Consequently an S-orbit of norm-8 vectors and its image in $L/2L$ have sizes in ratio $2:1$ ($-1 \in S$ acts trivially on $L/2L$), and since different canonical forms mean different orbits, $|K_8|$ equals the number of S-orbits.

C.4 Reduction of $(10, 6)$ to $(6, 10)$

If $v \equiv x\tau$ then $v\tau \equiv x\tau^2 = -5x \equiv x \pmod{2L}$. So (x, v) is a collision of type $(10, 6)$ iff (v, x) is one of type $(6, 10)$.

C.5 Orbit decomposition of $L_{56}(6)$

$632 \cdot 24\,360 + 24 \cdot 8\,120 = 15\,395\,520 + 194\,880 = 15\,590\,400$, and this is the only decomposition into 656 orbits with sizes in $\{8\,120, 24\,360\}$ summing to $|L_{56}(6)|$.

Appendix D. Data, formats and reproduction

Table D1. Data files (as distributed in [Repo], `data/`).

file	content	SHA-256 (prefix)
`L56_gram.txt`	Gram matrix G of L_{56} (56×56)	`4d13bd51…`
`sl2_29_generators.txt`	g_1, g_2 (orders 5, 7) generating $\mathrm{SL}_2(29)$	`97f88443…`
`tau.txt`	τ (56×56), $\tau^2 = -5$	`30b6fa54…`
`R112_gram_LLL.txt`	LLL-reduced Gram matrix of L_{112} (112×112)	`106ce481…`

Full SHA-256 hashes:

```
4d13bd51fcf3cb8c5706bde45828713d7eeb4d18a37e8e9d84fb2700f4bef2f0
97f88443de1fd1fdeb010ecb668dbbb03a8e0e989a18c0ad85a751687012ad90
30b6fa54f7abc2938a8f5ce29237bb92a4224f5418d3cae85183c0a89803a0f1
106ce481d60fac9bbbe211da1ba15c68610cd8dbcbee1fe0e110435f9a50083d
```

Conventions. Row vectors; a matrix g acts by $x \mapsto xg$; isometries satisfy $gGg^T = G$; $\tau G \tau^T = 5G$. The file `R112_gram_LLL.txt` is PARI's `qflllgram` reduction of (3.1).

Specification without the files. The defining matrices (56×56 and 112×112) are not printed in this paper; they are the files of Table D1. By §7.2, L_{56} is $\mathrm{SL}_2(29)$-equivariantly isometric to $L_{56,2}\big(\widetilde{M}\big)$ of [Neb98], and in those coordinates τ is right multiplication by an element of $\widetilde{M}$ of reduced norm 5 and reduced trace 0, given in §7.2. Together with (1.2), this specifies L_{112} up to isometry independently of the data files. The files remain the reference: every verification program reads them and checks their hashes.

Intermediate data (orbit representatives of $L_{56}(6)$ and $L_{56}(8)$, odd and even-coset certificates, witnesses of norm 8 for B) are distributed as verification data in `certificates/` of [Repo]; their SHA-256 hashes are listed in `certificates/SHA256SUMS`.

Elementary checks (a few seconds in any computer algebra system): (3.1) has even diagonal and determinant 1; $\tau^2 = -5$, $\tau G \tau^T = 5G$; $g_i G g_i^T = G$, $g_i \tau = \tau g_i$. Checking one odd-coset certificate: (a) $w_0 - r\tau$ has even coordinates, (b) $\langle w_0, w_0 \rangle \in \{14, 18\}$, (c) $|\langle w_0, r' g \rangle| \le 6$ for all 656 representatives r' and all 12 180 elements g.

Code. Every computation used in this paper is listed in Table D2 with the code that performs it; except for supplementary computations explicitly marked unpublished, the paths in Table D2 are those of the public repository [Repo]. The text refers to the rows of Table D2 as [C1], [C2], ….

Checking the computational facts from the supplied data takes about 1.2 CPU hours by the fastest route (route A of Supplement Table S7; the script `code/verify/proof.sh`). Alternative routes and additional tests are described in Supplement §S3. The identifier R112 in filenames is an earlier name for L_{112}.

Table D2. Computations and code. References such as [C1] in the text point to the rows of this table.

ID	computation and code (public-repository paths unless marked otherwise)	time
C1	Extremal theta series of L and L_{112}; Siegel means $\Sigma(n)$; Siegel orbit excess (exact rational arithmetic) `code/verify/v2_theta.py`	1 s

C2	Basic facts F0, F4: (3.1) is even with determinant 1; g_1, g_2 preserve G, commute with τ and generate a group of order 24 360 isomorphic to $\mathrm{SL}_2(29)$; $\tau^2 = -5$, $\tau G \tau^T = 5G$; the centralizer has order 12; the 24 elements with $\tau^2 = -5$ form 2 classes
	`code/verify/v0_inputs.py`
	`code/supplementary/gluings.py`
	`code/produce/p1_centralizer.py`
		10 min

C3	Gluings: the commutant of $\mathrm{SL}_2(29)$ on $L/2L$ is 8-dimensional (the image of M'); 60 invertible parity-preserving maps; their classes (Proposition 3.3, Table 2); forced collisions
	`code/supplementary/gluings.py`
	`code/supplementary/nebe/gluings_m.py`
		min

C4	Gluings A and B: minimum 8 and exactly 292 320 vectors of norm 8; invariants of A and B compared
	`code/supplementary/gluing_b.py`
	`code/supplementary/nebe/gluings_m.py`
		1.2 h

C5	F1: $\min L = 6$ by Fincke–Pohst enumeration with exact rational LDL^T and outward-rounded interval	34 min

ID	computation and code (public-repository paths unless marked otherwise)	time

	arithmetic (BKZ-40 basis, 8 454 837 756 nodes); controls E_8, Leech, Leech $\perp$ Leech $\perp E_8$
	`code/verify/v1_min_L.py`
	`code/verify/enum_interval.c`
	`code/produce/p6_reduction.py`	

C6	F1 by reduction to $L_{56,2}(M)$: no class of a norm-8 vector of $L_{56,2}(M)$ lies in the invariant subspace W (linear algebra over $\mathbb{F}_2$); positive control
	`code/supplementary/nebe/reduction_to_nebe.py`
	`code/supplementary/nebe/lattices.py`
	`code/supplementary/nebe/tables.py`
		27 s

C7	F1 by floating-point enumeration (fplll on two bases, PARI `qfminim`); supplementary `code/supplementary/float_enumerations.py`	51 min–2.3 h

C8	F2: the 656 orbit representatives of $L(6)$; distinct orbits whose sizes add up to 15 590 400; spherical design identities of degree 2, 4, 6
	`code/produce/p2_minimal_orbits.py`
	`code/verify/v3_minimal_orbits.py`
	`code/verify/v8_design_identities.py`
		$\approx$ 25 min

C9	F3: orbit representatives of $L(8)$ by marking sums of two minimal vectors; 1 517 130 distinct orbits whose sizes add up to 36 957 286 800; design identities
	`code/produce/p3_norm8_orbits.py`
	`code/verify/v4_norm8_orbits.py`
	`code/verify/v8_design_identities.py`
		1.2 h + 9 min

C10	F5a: odd-class certificates for the 656 classes $r\tau + 2L$ (norms 14, 18; inner products with all minimal vectors at most 6), and their re-check by separately written code
	`code/produce/p4_odd_descent.py`
	`code/verify/v5_odd_bounding.py`
	`code/crosscheck/second_impl/odd_classes.py`
		$\approx$ 10 min

C11	F5b by canonical forms on $L/2L$: the table K_8 and the look-up for all 1 527 984 pairs, in two implementations
	`code/verify/v6_canonical_lookup.py`
	`code/crosscheck/second_impl/even_classes.py`
		20 min–2 h

C12	Closest-vector enumerations (fplll, floating point) for the classes of F5a and F5b; supplementary `code/supplementary/float_enumerations.py`	2.7 h + 4 $\times$ 9 h

C13	F5b by even-class certificates for all 1 527 984 pairs (norms 12, 16, 20; inner products with all minimal vectors at most 6), and their check
	`code/produce/p5_even_descent.py`
	`code/verify/v7_even_bounding.py`
		6 $\times$ 31 h + 7.9 h

C14	Automorphisms preserving the standard frame; the halving of the pairs in route B `code/supplementary/frame_stabilizer.py`	1 s

ID	computation and code (public-repository paths unless marked otherwise)	time
C15	Route S: random sample of certificates re-checked with independent integer code `code/supplementary/sample_check.py`	8 min
C16	Gaussian estimate of the number of enumeration nodes for L_{112} itself (LLL, BKZ-20, BKZ-30 bases) `code/supplementary/node_estimate.py`	3 s
C17	Frame-change test: all 864 frames of Λ give the same class `code/supplementary/frames.py`	7 min
C18	Fixed sublattices of elements of order 3, 5, 7 and their exact minima; BKZ on L_{112} (tests that cannot separate B from C) `code/supplementary/fixed_sublattices.py`	40 s
C19	Support modules of the norm-8 vectors of B (Supplement Table S9) `code/supplementary/anatomy_b.py`	2 s

C20	Supplement §S4: rank-2 support modules (free and non-free), grouped into 97 invariant types, 63 dangerous. Both enumerations agree on 58 dangerous types after regrouping by the same coarser invariant
	`code/supplementary/route_t/classify.py`
	`code/supplementary/route_t/recount.py`
	`code/supplementary/route_t/hermitian.py`
		1–2 min each

C21	Supplement §S4: three searches modulo 2. All 3 936 certificates for test (3) checked in integers. Test (2): floating-point enumeration of all 763 992 classes; integer certificates for part of them (`certificates/route_t_search2.npz`, to be replaced when complete)
	`code/supplementary/route_t/certify.py`
	`code/supplementary/route_t/check.py`
	`code/supplementary/route_t/float_cvp.py`
	`code/supplementary/route_t/common.py`
		1.6 h + 3 × 13 h (CVP); (3) の代表元 25 min

C22	Lean formalization of the deduction from F1–F5 to Theorem A, and its check (no `sorry`; standard axioms only) `lean/EEUL/L112/Chain.lean`	min

C23	Identification of L with Nebe's $L_{56,2}(\widetilde{M})$ built from the definition; τ as right multiplication by an element of reduced norm 5
	`code/supplementary/nebe/identify_nebe.py`
	`code/supplementary/nebe/lattices.py`
	`code/supplementary/nebe/cycloquat.py`
		≈ 40 min

C24	The centralizer of $\mathrm{SL}_2(29)$ in $\mathrm{Aut}(L_{112})$ (order 48) and an automorphism of L_{112} inducing the outer automorphism `code/supplementary/automorphisms/aut_normalizer.py`	15 min

ID	computation and code (public-repository paths unless marked otherwise)	time

C25	Finite group facts G-F1–G-F5 used in Appendix F (GAP), and the diagonal case; further checks G-F6–G-F12 are retained in the code
	`code/supplementary/automorphisms/group_facts.g`
	`code/supplementary/automorphisms/out_facts.g`
	`code/supplementary/automorphisms/aut_diagonal.py`
		min

C26	Structure of $\mathrm{Aut}(L_{112})$: order of the generated group, non-split extension, quotients, Molien series; the $\mathbb{Z}\left[\sqrt{-29}\right]$- and $\mathcal{O}_K$-structures
	`code/supplementary/automorphisms/aut_structure.py`
	`code/supplementary/automorphisms/aut_character.py`
	`code/supplementary/automorphisms/endomorphisms.py`
		≈ 45 min

C27	Independent re-check of all certificates and of F1 by separately written code (integer arithmetic; float32 products only with proved bounds on all partial sums)
	`code/crosscheck/all_data/check_all.py`
	`code/crosscheck/all_data/min_L.py`
	`code/crosscheck/all_data/min_L_enum.c`
		≈ 10 h

C28	Gluings A and B are isometric: an explicit basis change T with $TG_BT^\top = G_A$, checked in integer arithmetic ($\det T = \pm 1$) `code/supplementary/isometry_ab.py`	find 8 min; check 2 s
C29	The $\mathrm{SL}_2(29)$-invariant even unimodular lattices of dimension 112 (`local`, `mass`, `enum`): the image of the group ring is maximal at $p = 2, 3, 5, 7, 29$ ($\mathbb{F}_p$-algebras of dimension 392), so every such lattice is $L_{56,2}(M) \otimes_M \Lambda$; $\zeta_K(-1) = 1/2$, $\zeta_K(-3) = 157/20$ ($K = \mathbb{Q}\left(\sqrt{29}\right)$, generalized Bernoulli numbers) and the mass $157/640$; Kneser neighbours at the prime of norm 5 (fixed random seed) reach the mass with 8 classes, $\lvert U(\Lambda)\rvert = 1152, 12, 288, 24, 24, 20, 288, 48$ (PARI `qfauto`) `code/supplementary/genus/genus.py`	local 20 min; mass 1 s; enum 16 min
C30	The 8 lattices $X_j = L_{56,2}(\mathfrak{M}) \otimes_{\mathfrak{M}} \Lambda_j$ (`build`; Gram matrices and bases in the coordinates of (3.1) in the verification data), their mod-2 frames (`frames`), and vectors of norm 8 in classes 3, 4, 5 from fixed sublattices (`witness`) `code/supplementary/genus/genus.py`	build 4 min; frames 12 min; witness 2 min
C31	Explicit $\mathrm{SL}_2(29)$-equivariant isometries class $7 \to L_{112}$ and class $3 \to$ gluing A, and a recount of $\lvert U(\Lambda_j)\rvert$ as equivariant automorphisms (`identify`); integer-arithmetic check, numpy only (`check`): each G_X even with determinant 1, g_1, g_2 act on X_j, the vectors of norm 8, the isometries ($\det T = \pm 1$), and $\sum_j 1/\lvert U(\Lambda_j)\rvert = 157/640$ `code/supplementary/genus/genus.py`	identify 13 min; check 80 s

Appendix E. Pseudocode

```
MINIMAL-ORBITS(L, S):
  candidates <- initial minimal vectors; reps <- []; seen <- {}; total <- 0
  while total < 15590400:
    r <- a candidate outside seen; O <- {r*g : g in S}
    append r to reps; seen <- seen union O; total <- total + size(O)
    for s in O with <r,s> = 3: add r-s to candidates
    if no unused candidates remain: enlarge the initial candidate list
  check distinct orbits and total = 15590400
  return reps

NORM8-ORBITS(L, S_star, minimal_reps):
  for r_i in minimal_reps and s in L(6) with <r_i,s> = -2:
    x <- r_i+s
    if x is marked: continue
    find and mark all marked elements of x*S_star using <x*u,r_j*g> = +/-4
    record x and size(S_star)/size(stabilizer(x))
  check distinct orbits and total orbit size = 36957286800

CANONICAL(c, S):
  return min { pack56(c*g mod 2) : g in S/{+1,-1} }

TEST88(pairs, K8, tau):
  for (x_i,u) in pairs:
    if CANONICAL(x_i*u*tau, S) in K8: report collision

COSET-REPRESENTATIVE(w):
  repeat:
    m <- max { <w,u> : u in L(6) }
    if m <= 6: return w
    choose u with <w,u> = m; w <- w-2*u
  # Check 5.4 also verifies parity and the allowed final norm.
```

Appendix F. Proof that $\mathrm{SL}_2(29)$ is normal in $\mathrm{Aut}(L_{112})$

Write $A = \mathrm{Aut}(L_{112})$, $S \cong \mathrm{SL}_2(29)$, $V = L_{112} \otimes \mathbb{Q}$, so $V|_S \cong V_0^2$ with $V_0 = L_{56} \otimes \mathbb{Q}$ irreducible of dimension 56 and $V \otimes \mathbb{C}|_S = 4\psi + 4\psi'$, where ψ, ψ' are the two faithful characters of degree 14. The generalized Fitting subgroup $F^*(A) = F(A)E(A)$ satisfies $C_A(F^*(A)) \le F^*(A)$ (Bender). We show (F.1) that every normal subgroup of A has at most two isotypic components on V; (F.2, F.3) that $F(A)$ centralizes S; and (F.4) that $E(A) = S$. Sakamoto [Sak26, App. A] used primitivity of the action, which fails here (there are two isotypic components a priori); F.1 replaces it, and F.3 replaces the use of "normal abelian subgroups are cyclic" by a character-degree argument. The finite facts G-F1–G-F5 below were checked with GAP [C25].

Finite facts. G-F1: the proper subgroups of $\mathrm{PSL}_2(29)$ of index ≤ 112 are $29:14$ (index 30) and $29:7$ (index 60). G-F2: every proper subgroup of $\mathrm{SL}_2(29)$ has index ≥ 30. G-F3: the irreducible characters of the preimages in S of $29:14$, $29:7$ have degrees $1, 14$ and $1, 7$; ψ, ψ' restricted to a subgroup U of order 29 do not contain the trivial character. G-F4: the only maximal subgroup of Ru of order divisible by 29 is $L_2(29)$; both characters of degree 28 of 2. Ru restrict irreducibly to $2.L_2(29)$, for each of the 6 possible class fusions. G-F5: $|\mathrm{SL}_2(29)|$ divides neither $|\mathrm{SL}_2(59)|$ nor $|\mathrm{SL}_2(173)|$.

F.1 At most two isotypic components

Let $N \trianglelefteq A$ and let $V_1, ..., V_b$ be the isotypic components of $V|_N$ over $\mathbb{Q}$. A permutes them (Clifford), hence so does S, through $\mathrm{PSL}_2(29)$ since -1 is scalar. If the permutation action were nontrivial, a nontrivial orbit would have a stabilizer of index ≤ 112, so of index 30 or 60 (G-F1), and its components would have dimension ≤ 3. The sum of the orbit would be induced from a representation θ of dimension ≤ 3 of the preimage J of the stabilizer; by G-F3, θ is a sum of linear characters, trivial on $U \leq [J, J]$. A common constituent of θ and $\psi|_J$ would give a trivial constituent of $\psi|_U$, contradicting G-F3; so by Frobenius reciprocity $\mathrm{Ind}_J^S \theta$ contains neither ψ nor ψ', which is impossible inside $V|_S$. Hence S fixes every V_i, each V_i is an S-submodule of V_0^2 of dimension 56 or 112, and $\boldsymbol{b \leq 2}$.

F.2 Odd primes

Let $P_0 \trianglelefteq A$ be a p-group, p odd, with kernel K_j on V_j. P_0/K_j has a faithful irreducible rational representation U_j; for a constituent χ of $U_j \otimes \mathbb{C}$, $\dim U_j = [\mathbb{Q}(\chi) : \mathbb{Q}]\chi(1)$ (Schur index 1, Roquette), and $\mathbb{Q}(\chi) \supseteq \mathbb{Q}(\zeta_{p^f})$ because $Z(P_0/K_j) \cong C_{p^f}$ ($f \geq 1$) acts through a faithful linear character. So $\dim U_j = (p-1)p^e\chi(1)$ divides 56 or 112, which forces $p^e\chi(1) = 1$ and $p - 1 \mid 112$, i.e. $p \in \{3, 5, 17, 29, 113\}$ and $P_0/K_j \cong C_p$. Thus $P_0 \hookrightarrow C_p \times C_p$. For $p \neq 29$, $29 \nmid |\mathrm{GL}_2(p)|$, so the perfect group S acts trivially on P_0. For $p = 29$: if P_0 is cyclic, $\mathrm{Aut}(P_0)$ is abelian; if $P_0 \cong C_{29}^2$ then $b = 2$ and S fixes K_1 and K_2 (F.1), so its image in $\mathrm{GL}_2(29)$ lies in a split torus, which is abelian. In all cases $[P_0, S] = 1$.

F.3 The prime 2

Let $P = O_2(A)$. S is generated by its elements of order 29, so it suffices to show that such an element y centralizes P. Let D_0 be a critical subgroup of P in the sense of Thompson: D_0 is characteristic in P, of class ≤ 2, and every nontrivial automorphism of P of odd order acts nontrivially on D_0. Since $D_0 \trianglelefteq A$, F.1 applies: $V|_{D_0}$ has isotypic components V_1 (and V_2), fixed by y; let K_j be the kernel on V_j, so $K_1 \cap K_2 = 1$ and y acts on $Q_j = D_0/K_j$. A constituent χ of $V_j \otimes \mathbb{C}$ is a faithful irreducible character of Q_j (Galois conjugates have the same kernel) with $\chi(1) \leq 112$.

Lemma. If Q has class ≤ 2 and $\chi \in \mathrm{Irr}(Q)$ is faithful, then $\chi(1)^2 = |Q : Z(Q)|$. *Proof.* For $g \notin Z(Q)$ pick h with $z = [g, h] \neq 1$; $z \in Z(Q)$ acts as a scalar $\lambda(z) \neq 1$, and $\chi(g) = \chi(h^{-1}gh) = \chi(gz) = \lambda(z)\chi(g)$, so $\chi(g) = 0$. The orthogonality relation $\sum_g |\chi(g)|^2 = |Q|$ gives $\chi(1)^2|Z(Q)| = |Q|$. ■

Hence $|Q_j : Z(Q_j)| \leq 112^2 < 2^{14}$, and $Z(Q_j)$ is cyclic. The element y acts trivially on $Z(Q_j)$ (the automorphism group of a cyclic 2-group is a 2-group) and on the abelian group $Q_j/Z(Q_j)$: otherwise it would act nontrivially on its Frattini quotient (Burnside), an $\mathbb{F}_2$-space of dimension ≤ 13, whereas nontrivial irreducible $\mathbb{F}_2[C_{29}]$-modules have dimension 28 (the order of 2 modulo 29). Writing $x^y = x\,\zeta(x)$ with $\zeta : Q_j \to Z(Q_j)$ a homomorphism, $x = x^{y^{29}} = x\,\zeta(x)^{29}$ forces $\zeta = 1$. So $[D_0, y] \leq K_1 \cap K_2 = 1$, and by the property of D_0, y centralizes P.

F.4 The layer

By F.2–F.3, $F(A) \leq C_A(S)$. Let $E(A) = Q_1 \cdots Q_r$ be the product of the components (commuting quasisimple subnormal subgroups).

(a) *S normalizes every Q_i.* By F.1 for $E(A)$, $V|_{E(A)}$ has at most two isotypic components; the complex constituents of the j-th are Galois conjugates of $X_j = \bigotimes_i X_{j,i}$. Each Q_i acts nontrivially, nontrivial irreducible representations of quasisimple groups have degree ≥ 2, and $2^7 > 112$, so at most 6 of the Q_i act nontrivially on each of the at most two components; hence $r \leq 12$. By G-F2, S cannot permute the Q_i nontrivially.

(b) *S induces inner automorphisms on each Q_i.* By the Schreier conjecture, a consequence of the classification theorem, the outer automorphism group of the simple group $\frac{Q_i}{Z(Q_i)}$ is solvable. Automorphisms of a quasisimple group act faithfully on its central quotient, so $\mathrm{Out}(Q_i)$ is solvable as well. Since S is perfect, its image in $\mathrm{Out}(Q_i)$ is trivial. Thus $S \le E(A)C_{A(E(A))}$.

(b′) $S \le E(A)$. Put $T = SE(A) \cap C_A(E(A))$; then $SE(A) = E(A)T$ and $T/Z(E(A)) \cong SE(A)/E(A)$ is perfect. As S and $E(A)$ centralize $F(A)$, $T \le C_A(F^*(A)) \le Z(F^*(A))$ is abelian. So $T \le E(A)$ and $S \le E(A)$.

(c) *S lies in a single component.* Let I be the set of i for which the projection of S to $Q_i/Z(Q_i)$ is nontrivial. For $i \in I$, Q_i contains a quotient of $\mathrm{SL}_2(29)$, so its nontrivial irreducible representations have degree ≥ 14; as $14^2 > 112$, each X_j involves at most one $i \in I$, so $|I| \le 2$. If $|I| = 2$, then $b = 2$, Q_1 acts only on V_1 and Q_2 only on V_2; writing $s = zq_1q_2$ ($z \in Z(E(A))$, $q_i \in Q_i$) gives $[s,t] = [q_1, q_1'][q_2, q_2']$ with $[q_1, q_1'] \in A$ acting as $[s,t]$ on V_1 and trivially on V_2. Since S is perfect, $s \mapsto (s|_{V_1}, \mathrm{id}_{V_2})$ maps S into A; the image c of -1 is an involution in $C_A(S)$ for which $g \mapsto \frac{1-c}{2}g + \frac{1+c}{2}$ preserves L_{112}. A direct check of the 6 involutions $\ne \pm 1$ of the group of order 48 (Theorem B(a)) shows that none does [C25]. So $|I| = 1$ and $S = [S,S] \le [Q_1, Q_1] = Q_1$.

(d) *$E(A) = Q_1$.* Any other component centralizes $Q_1 \supseteq S$ and hence lies in $C_A(S)$ of order 48; but quasisimple groups have order ≥ 60.

(e) *$Q_1 = S$.* Let X be a constituent of $V \otimes \mathbb{C}|_{Q_1}$. Since $-1 \in S$ acts as -1, $\bar{Q} = Q_1/\ker X$ is quasisimple with center of even order, contains S faithfully, and X is a faithful irreducible representation of degree ≤ 112 whose restriction to S is a sum of ψ, ψ'. By the main theorem of [HM01] (with the corrigenda [HM02]), such a $\bar{Q}$ in characteristic 0 with $29 \mid |\bar{Q}|$ is one of the following. From [HM01, Table 3], only Ru, 2. Ru, J_4 have order divisible by 29 (we checked the orders of all simple groups occurring in rows of degree ≤ 112), and only 2. Ru (degree 28) occurs in characteristic 0 with even center. From [HM01, Table 2], A_n has trivial center, and $2.L_2(q)$ with $29 \mid q(q^2-1)$ and minimal degree $(q \mp 1)/2 \le 112$ gives $q \in \{29, 59, 173\}$, of which 59 and 173 are excluded by G-F5. If $S \le 2.\mathrm{Ru}$, the image of S lies in the maximal subgroup $L_2(29)$ (G-F4), so $S = 2.L_2(29)$, and $X|_S$ would be irreducible of degree 28 (G-F4), not a sum of characters of degree 14. Hence $\bar{Q} \cong \mathrm{SL}_2(29)$, $Q_1 = S \ker X$ with $\ker X$ central, and $Q_1 = [Q_1, Q_1] = S$.

Thus $E(A) = S$, which is characteristic in A; so $S \trianglelefteq A$, $A = N_A(S)$ and $|A| = 1\,169\,280$ by Theorem B(b). □

What the proof relies on. The finite computations G-F1–G-F5, Theorem B(a), (b) and the diagonal check [C25] (reproducible in minutes); the main theorem of Hiss–Malle (whose tables were scanned mechanically for all rows of degree ≤ 112); standard textbook facts of finite group theory (Bender, Thompson's critical subgroup, Burnside, Clifford, Roquette).

For the standard finite-group results on generalized Fitting subgroups, critical subgroups and Clifford theory, see [Asc00].

Appendix G. Comparison with known constructions

Table G1. Selected constructions and counting results for extremal even unimodular lattices; historical sources are [Sak26, §1] and [NS].

dimension n	$\mu(n)$	examples or counting results	sources
8	2	E_8 (unique)	classical
16	2	$E_8 \perp E_8$, D_{16}^+ (all)	classical
24	4	Leech lattice (unique)	[SPLAG]
32	4	more than 10^7	mass formula by root system [Kin03, Cor. 5]
40	4	more than 10^{51}	mass formula and Siegel's theorem [SSP99]; [Sak26, §1]
48	6	at least 4	[Sak26, §1]; [NS]
56	6	Nebe's constructions [Neb98]	[Sak26, §1]; [NS]; [Neb98]
64	6	constructions of Quebbemann and Shimada	[Que84]; [Sak26, §1]; [NS]
72	8	at least 1 (Γ_{72})	Nebe 2010 [Neb12]
80	8	at least 4	Bachoc–Nebe 1998 [BN98]; [SW10]; [Wat12]
88	8	at least 2	Sakamoto 2026 [Sak26] (preprint)
96	10	none known	open
104	10	none known	open
112	**10**	$\boldsymbol{L_{112}}$ **(this paper)**	this paper
120	12	none known	open